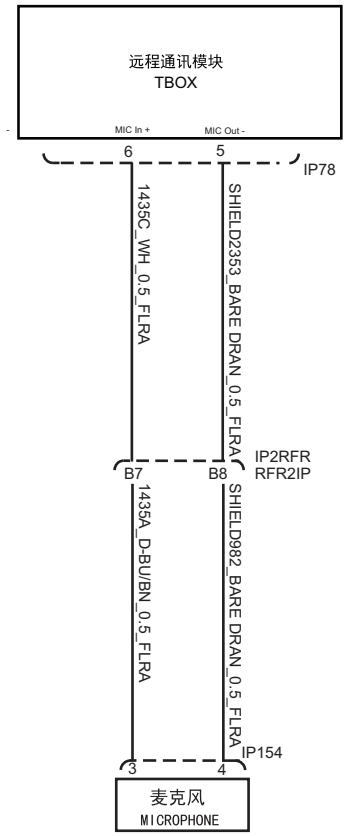
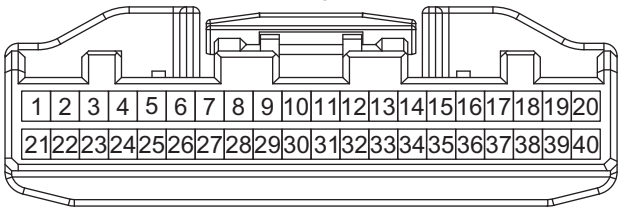
B1533-11、 B1533-13、 B1534-11、 B1534-13

Description of fault diagnosis codes
B1533-11：Microphone Input short to Ground
B1533-13：Microphone Input Open Ground
B1534-11：Microphone Out short to Ground
B1534-13：Microphone Out Open Ground
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • TBOX fault • Microphone fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1533-11
B1533-13
B1534-11
B1534-13
To set the effect of a fault code condition
1 2
Description of circuit diagram
T-BOX module monitors microphone signals by connecting wire harnesses.

Circuit diagram



IP78



IP154

EV901105

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of microphone,TBOX for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the TBOX with diagnostic apparatus. 4 Is there any other fault code except for B1533-11、B1533-13、 B1534-11、 B1534-13?
		YES → Refer to:DTC Summary List (TBOX).
NO ↓		
Step 3	Check the power supply circuit of microphone circuit.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP154 of microphone. 3 Turn the ignition switch to ON. 4 Measure the voltage between the harness terminal of microphone and ground. Measuring circuit: voltage between terminal 3 on plug IP154 and ground. **Standard value:** 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check the circuit of Microphone (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP78 of TBOX. 3 Measure the resistance between the harness terminal of microphone and of TBOX. Measuring circuit: resistance between terminal 3 on plug IP154 and terminal 6 on plug IP78. Measuring circuit: resistance between terminal 4 on plug IP154 and terminal 5 on plug IP78. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the circuit of Microphone (open circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of microphone and ground. Measuring circuit: resistance between terminal 3 on plug IP154 and ground. Measuring circuit: resistance between terminal 4 on plug IP154 and ground. **Standard value: ∞** 3 Measure the resistance between the harness terminal of microphone and ground. Measuring circuit: voltage between terminal 3 on plug IP154 and ground. Measuring circuit: voltage between terminal 4 on plug IP154 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the microphone (parts).
	1 Measure the resistance of microphone terminal directly. Measuring circuit: resistance between terminal 3 and terminal 4 on plug IP154. **Standard value:** 2 Measure the resistance between the microphone terminal and the housing. Measuring circuit: resistance between terminal 3 on plug IP154 and the housing. Measuring circuit: resistance between terminal 4 on plug IP154 and the housing. **Standard value: ∞** 3 Check whether the result is normal or not?
	NO → Replace the ambient temperature sensor.
YES ↓	
Step 7	Test whether TBOX is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug IP78 of TBOX. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the TBOX plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the TBOX.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	